Classical Waves: A Universal Structure

This lesson is intended as part of an upper-level undergraduate course in Waves and/or Classical Physics. Prerequisites: Calculus I, II, and III; Linear Algebra; Differential Equations (and optionally Partial Differential Equations); Math physics I; E&M I and II; Classical Mechanics; Data Analysis/Science/Programming

In this lecture, we will develop the idea of waves as a unifying physical concept, beginning with classical mechanical systems, progressing through electromagnetic fields, and culminating in gravitational waves. Although the physical interpretation of a wave differs across these systems, we will see that the mathematical structure governing wave propagation is remarkably universal. Be sure to finish the 5 homework exercises here along with the problems assigned from the textbook.

Part I
Lecture

1 The Wave Equation

1.1 Physical Assumptions and Domain of Validity

We begin with classical mechanical waves, such as waves on strings, sound waves in fluids, or surface waves on water. The wave equation describing these systems arises only under a specific set of assumptions.

First, we assume that the system is close to equilibrium and that disturbances are small. This allows us to linearize the equations of motion and neglect higher-order terms. Also, we assume that the medium is continuous, homogeneous, and isotropic on the length scales of interest; we assume locality: the restoring force at a point depends only on nearby displacements; and we assume time-translation invariance (the physical laws governing the system do not change with time). These assumptions imply that *linear superposition* holds, allowing us to decompose complex waveforms into simpler components. When any of these assumptions fail—such as in strongly nonlinear waves, shocks, or highly inhomogeneous media—the wave equation may no longer apply.

1.2 Mechanical Waves

Consider a transverse wave on a stretched string. Let $y(x,t)$ denote the transverse displacement from equilibrium. Applying Newton's second law to an infinitesimal segment of the string, one finds

$$\mu \frac{\partial^2 y}{\partial t^2} = T \frac{\partial^2 y}{\partial x^2}, \tag{1}$$

where μ is the mass per unit length and T is the tension. This leads to the one-dimensional wave equation

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}, \quad v = \sqrt{\frac{T}{\mu}}. \tag{2}$$

More generally, this equation applies to longitudinal and transverse waves. In all cases, the key feature is that the wave equation relates second derivatives in space and time.

Homework Exercise # 1: compare this with the heat/diffusion equation.

1.3 The Three-Dimensional Wave Equation

In three spatial dimensions, the wave equation for a scalar field $\psi(\mathbf{x}, t)$ takes the form

$$\left(\nabla^2 - \frac{1}{v^2}\frac{\partial^2}{\partial t^2}\right)\psi(\mathbf{x}, t) = 0. \tag{3}$$

This equation admits a wide range of solutions, but physical solutions depend crucially on boundary and initial conditions. For example, standing waves arise in bounded domains, while traveling waves are appropriate in unbounded space.

1.4 Methods of Solution

Several complementary methods exist for obtaining analytic solutions to the wave equation. The first method you typically encounter in your PDE course is the method of separation of variables, where one can algebraically rewrite an equation so that each variable occurs on a different side of the equation (or if more than two dimensions, occurs in independent functions).

In bounded domains, Sturm–Liouville theory allows the solution to be expressed as an expansion in orthogonal eigenfunctions determined by boundary conditions. For linear homogeneous PDEs, this produces linearly independent ODEs for time and space, where the spatial parts become a Sturm-Liouville ODE: $\frac{d}{dx}[p(x)\frac{dy}{dx}] + q(x)y = -\lambda w(x)y$. These separated ODEs are typically eigenvalue problems, which are designed to generate a set of linearly independent eigenfunctions (e.g., sines and cosines) that can be combined to form the general solution. See your mathematical physics textbook, e.g. pg.'s 381 and 414 of Arfken, Weber, and Harris (7th Ed.).

In Fourier analysis, any localized disturbance can be decomposed into plane waves, each propagating at speed v. It is particularly useful for linear PDEs on infinite or periodic domains (using Fourier Transforms: transforming complex spatial derivatives into simpler algebraic equations or ODEs in the frequency domain, solving them, and applying the inverse transform), or bounded domains (using Fourier Series with separation of variables).

The Green's function approach models propagation from localized sources by solving for the causal "impulse response" of the system. In this method, the wave equation is first solved for a singular point source (a Dirac delta function), resulting in the retarded Green's function, and the complete solution for any arbitrary source distribution is then constructed by integrating this response across the entire history and spatial extent of the source.

Note on linearity: specific nonlinear PDEs can be solved analytically, but no general method exists for all cases. Analytical solutions usually require special equations, and are rare and mostly restricted to specific, specialized types. As we will see later in the case of Einsteinian gravity, this necessitates the use of numerical solution techniques.

Caveat: The wave equation assumes the disturbance exists for all time and space, which is an idealization. In real physical systems, wave packets have finite extent, and dispersion or dissipation may modify their evolution. The apparent universality of the wave equation therefore reflects both deep physical principles and carefully chosen approximations.

2 Electromagnetic Waves

2.1 Maxwell's Equations in Vacuum

Electromagnetic waves emerge naturally from Maxwell's equations. In your E&M course(s), you likely used them in their integral forms in order to solve high-symmetry problems in electrostatics,

magnetostatics, and basic induction. Here, we will use their differential forms.

In vacuum, these equations are

$$\nabla \cdot \mathbf{E} = 0, \quad \text{(Gauss's law for electricity)} \tag{4}$$

$$\nabla \cdot \mathbf{B} = 0, \quad \text{(Gauss's law for magnetism)} \tag{5}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \text{(Faraday's law)} \tag{6}$$

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad \text{(Ampère–Maxwell law)} \tag{7}$$

Here, $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields, while ϵ_0 and μ_0 are the permittivity and permeability of free space.

2.2 Derivation of the Electromagnetic Wave Equation

To derive the wave equation for the electric field, we take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}). \tag{8}$$

Using the vector identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ and Gauss's law for electricity, we find

$$-\nabla^2 \mathbf{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{9}$$

Rearranging yields the wave equation

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{E} = 0, \quad c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}, \tag{10}$$

where c is the speed of light in vacuum. Thus, light is an electromagnetic wave!

Homework Exercise # 2: derive the wave equation for the magnetic field $\mathbf{B}$ in vacuum.

2.3 Physical Interpretation

Unlike mechanical waves, electromagnetic waves do not require a material medium. Instead, the electric and magnetic fields sustain each other through mutual induction, allowing the wave to propagate through vacuum, which can be seen in Faraday's Law and Ampere-Maxwell Law where a time varying E or B field sources a B or E field, respectively. See here for a friendly animation. See the Falstad simulation for more details.

Maxwell's equations are linear, allowing for the methods from Sec. 1.4 to be used. For example, separation of variables is highly effective for problems with cylindrical, spherical, or Cartesian symmetry (e.g., waveguides, antennas, cavity resonators). Other techniques, such as the method of images (e.g., point charges near grounded conductors) or numerical approaches (e.g., finite element method or method of moments for dielectrics or wave-matter interactions) are also useful.

The development of these equations and the quest to understand them led to many breakthroughs in technology and physics: (1) new methods for telecommunication (i.e., transmission cables) and record keeping (i.e., electronics and electrical engineering); (2) the realization of the quantum realm due to issues with combining Maxwell's laws with thermodynamics (i.e., equipartition theorem) leading to the UV catastrophe; and (3) the emergent field of spectroscopy through the unification of electricity, magnetism, and optics as represented by the electromagnetic spectrum.

3 Gravitational Waves

Einsteinian gravity is the modern view of the gravitational interaction, known as the Theory of General Relativity (GR). Since a course on GR is not an expected prerequesite, I will provide a basic introduction to the concepts needed to understand an interesting prediction known as *gravitational waves* (GWs).

3.1 Crash Course in GR

Newton's theory of motion and gravity relied on the Galilean transformations that relate coordinate systems of observers in relative motion, which has known pathologies. In 1905, Einstein derived the improved Lorentz transformations from simple physical assumptions, or axioms, which formed the Theory of Special Relativity.

To obtain GR, Einstein combined Special Relativity and the equivalence principle, which states that gravitational and inertial mass are equivalent. This resulted in many interesting things, such as the abolition of absolute space and time - instead, space and time are aspects of a unified *spacetime*. In practice, this meant that the clocks of observers at different locations in spacetime can tick at different rates in the presence of gravity, which is modeled as the curvature of spacetime. For example, Global Positioning Systems rely on GR to accurately compute the location of a person on Earth's surface. John A. Wheeler famously explained, *"Spacetime tells matter how to move; matter tells spacetime how to curve."*

Question 1: Is gravitational free-fall an inertial or non-inertial motion?

In the Newtonian perspective, free-fall is the case where there is only the force of gravity acting on an object, so you have a free-body diagram with an arrow pointing downward, implying free-fall motion is non-inertial, while inertial motion requires an equal and opposing force to gravity.
In the Einsteinian perspective, free-fall is inertial motion because test particles follow trajectories governed by the inertial (i.e. geodesic) equation. Historically, this is why Einstein referred to GR as a unification of gravity and inertia. So Einstein kind of flipped gravity on its head :)

In GR, the curvature of four dimensional spacetime replaces the notion of the gravitational force vector from Newtonian gravity. This is mathematically captured by a matrix, known as the metric $g_{\mu\nu}$, which is a fundamental tool of differential geometry, and can be interpreted as the gravitational field. The metric, $g_{\mu\nu}$, is the solution to the Einstein field equations, which are a set of nonlinear, coupled PDEs. There are a few thousand known analytic solutions, perhaps most famously the black hole solutions, but generally these equations are highly nonlinear and very challenging to solve.

GR possesses gauge freedom because it is built upon general covariance, meaning physical laws are independent of the specific coordinate system (gauge) chosen to describe spacetime. This is a consequence of the symmetry group of GR being the group of diffeomorphisms. This has caused much confusion in the history of GR, and means that one must be careful to not confuse coordinate artifacts for physical features.

While early notions of GWs traced back to Oliver Heaviside and Henri Poincare around the turn of the 20th century, they are a unique aspect of GR. Einstein derived the first GWs model from GR by linearizing his field equations in the limit of asymptotic flatness.

Question 2: If the Sun disappeared instantly, would we know instantly?

In the Newtonian perspective, yes because information travels instantaneously, including the gravitational force.
In the Einsteinian perspective, no because nothing travels faster than the speed of light in vacuum.

It is important to note that the application of Newton's laws of motion and gravity to astronomical objects was important for testing their validity. In Newtonian gravity, two objects that are gravitationally bound to each other will orbit a common center of mass for infinity. One obtains Kepler's laws of planetary motion as a consequence of Newton's laws and angular momentum conservation.

In Einsteinian gravity, orbits decay due to the emission of GWs. That is, the gravitational energy of the orbit is transformed to the energy of the emitted GW. We will examine this in detail in the coding demo at the end of this lecture.

3.2 Linearized Einsteinian Gravity

In regions far from sources, where spacetime curvature is weak, we write the metric as a small perturbation about flat spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1\,, \tag{11}$$

where $\eta_{\mu\nu}$ is the Minkowski metric (ie flat spacetime, no gravity) and the subscripts are indices denoting the rows and columns of the matrix. To first order in $h_{\mu\nu}$, using the above metric perturbation turns the Einstein field equations into linear differential equations.

After choosing a suitable gauge condition, similar to the case in Maxwell's equation for the magnetic field B or magnetic vector potential A, the vacuum linearized Einstein equations are,

$$\Box \bar{h}_{\mu\nu} = 0, \tag{12}$$

where $\Box = \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$ is the d'Alembertian operator. For a full derivation of this, see pg. 267 of R. D'Inverno's book on Relativity.

Further gauge fixing leads the only two independent components of $h_{\mu\nu}$: the "plus" and "cross" polarization states of GWs. Thus GWs have two polarization states that specify the full wave, $h = h_+(t) + ih_x(t)$, compared to the case of EM waves which only require one polarization state.

3.3 Physical Interpretation

GWs are oscillations of spacetime itself, implying that GWs can propagate through vacuum and interact (extremely weakly) with matter.

The physical reality of GWs was famously debated in the 20th century. Einstein and Rosen argued that GWs do not exist in reality, claiming they were a pathology of general relativity. In the 1950s, Feynman's "sticky bead" argument claimed that GWs must carry energy, and a passing gravitational wave can do work on a system, producing observable effects. This idea is exploited by modern GW detectors to measure the tiny effects of GWs on test masses. Thus, in classcial GR, we can interpret GWs as the transport of gravitational energy, i.e., gravitational radiation. The simplest way to model GWs is in the linearized approximation to the Einstein field equations, where gravity is a field propagating on a flat background.

In interferometric detectors such as LIGO, GWs produce a fractional change in arm length known as the dimensionless strain,

$$h = \frac{\Delta L}{L}\,. \tag{13}$$

For a typical astrophysical source, $h \sim 10^{-21}$. For a 4 km interferometer, this corresponds to a displacement

$$\Delta L \sim 4 \times 10^{-18}\,\mathrm{m}\,, \tag{14}$$

which is approximately *one ten-thousandth the diameter of a proton* !!

Homework Exercise # 3: describe how the LIGO detectors are able to detect GWs; consider the terrestrial sources of noise that must be accounted to detect GWs, and consider the important technological aspects of the LIGO detectors.

3.4 Physical Sources

As gravity is a weak force, we need very energetic systems to produce observable GWs: the canonical examples are the coalescences of binary neutron stars (BNSs) and binary black-holes (BBHs).

The first evidence of GWs was indirect. It came from EM observations of the decay rate of the orbit of the first ever observed BNS (also known as a binary pulsar). The first direct evidence of GWs was reported[1] in 2016 by the LIGO consortium from the coalescence of a BBH, called GW150914, nearly a century after Einstein predicted their properties.

Generally, GWs arise from time-dependent, non-spherically symmetric mass distributions. Conservation laws forbid monopole and dipole gravitational radiation for isolated systems, *making the mass quadrupole moment the leading-order source of GWs*. In the weak-field, slow-motion limit, the GW strain amplitude is given by the quadrupole formula,

$$h_{ij}(t, \mathbf{x}) = \frac{2G}{c^4 r}\frac{d^2}{dt^2}Q_{ij}(t - r/c), \tag{15}$$

where r is the distance to the source and

$$Q_{ij} = \int \rho(\mathbf{x})\left(x_i x_j - \tfrac{1}{3}\delta_{ij}r^2\right)\, d^3x \tag{16}$$

is the reduced mass quadrupole moment of the source (this assumes the transverse-traceless gauge).

3.5 Post-Newtonian Inspiral Strain

While the quadrupole formula above encapsulates the dominant dependence of the GW, additional terms can be added to the quadrupole formula to better approximate the solution to the Einstein equations - this is known as the *post-Newtonian* approach. Here, we will only consider the leading (Newtonian) dependence.

For a compact binary in a quasi-circular orbit, the quadrupole formula yields gravitational radiation at twice the orbital frequency, and the observed strain takes the form

$$h(t) = \frac{4}{r}\left(\frac{G\mathcal{M}}{c^2}\right)^{5/3}(\pi f(t))^{2/3}\cos[\phi(t)]\,, \tag{17}$$

[1]The detection occurred in 2014, see Fig. 1 of the paper.

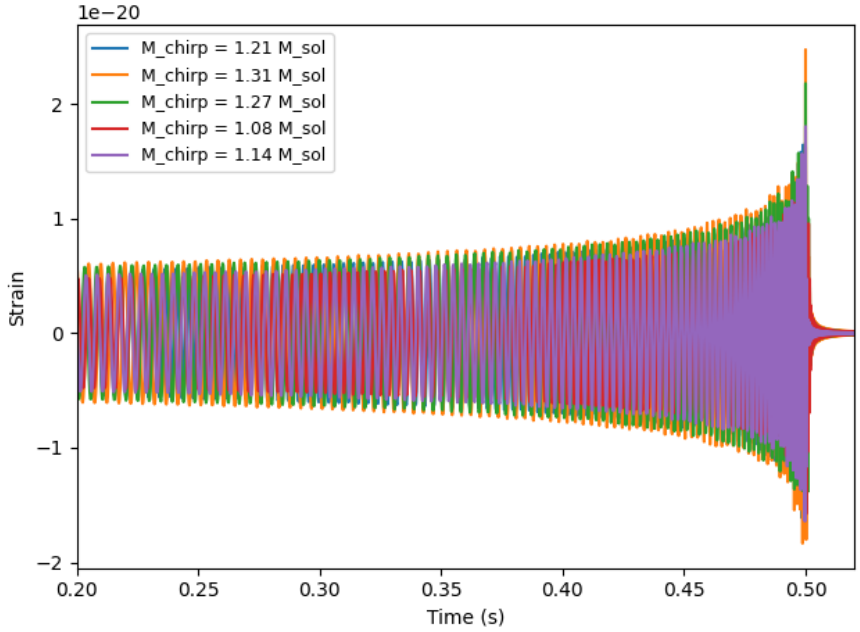


Figure 1: Five BNS waveforms corresponding to 5 different values of chirp mass $\mathcal{M}$.

where $\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1+m_2)^{1/5}}$ is the chirp mass, $f(t)$ is the instantaneous GW frequency, and $\phi(t)$ is the accumulated phase.

The frequency is governed by energy balance[2],

$$\frac{df}{dt} = \frac{96}{5}\pi^{8/3}\left(\frac{G\mathcal{M}}{c^3}\right)^{5/3} f^{11/3}, \tag{18}$$

which integrates to

$$f(t) = \frac{1}{\pi}\left[\frac{5}{256}\frac{1}{(t_c - t)}\left(\frac{G\mathcal{M}}{c^3}\right)^{-5/3}\right]^{3/8}, \tag{19}$$

where t_c is the coalescence time. Substituting this into the strain equation yields the characteristic "chirp": an oscillatory signal whose amplitude and frequency increase as the binary inspirals, providing a direct observational handle on the chirp mass and distance of the source:

$$h(t) = \frac{4}{r}\left(\frac{G\mathcal{M}}{c^2}\right)^{5/4}\left(\frac{5}{256}\right)^{1/4}(t_c - t)^{-1/4}\cos[\phi(t)]\,. \tag{20}$$

Figure 1 shows an example of the BNS strain plotted vs time for a few chirp masses. Note that the above formula is for the inspiral portion only, but we have plotted the full inspiral, merger, and ringdown using the IMRPhenomD_NRTidal model[3].

While the above equation captures the approximate inspiral waveform, ultimately, we need three different tools to model the full coalescence of a compact binary: post-Newtonian approximation (inspiral) + Numerical Relativity (merger) + Perturbative black hole (ringdown). Nowadays, there exist highly specialized waveform families. We will use them in our coding demo next.

Homework Exercise # 4: does the above waveform solve the (linearized Einstein) wave equation? (Hint: is it an exact or approximate solution?)

[2]See the classic paper by Damour (2001), and the textbook by E. Poisson called *Gravity*.
[3]See this paper.

Part II
Jupyter Notebook Demo

This notebook contrasts two conceptually distinct approaches to GW analysis: classical matched filtering, which is statistically optimal under well-defined noise assumptions, and machine-learning-based classification, which instead learns waveform morphology directly from data. The purpose is not to advocate machine learning as a replacement for matched filtering, but to use it as a controlled comparison that clarifies what physical information GW signals contain and how that information is extracted.

The notebook is organized into four parts. First, a faithful reproduction of the matched-filter signal-to-noise ratio (SNR) calculation for a real LIGO detection is presented. Second, a clean machine-learning classification problem is constructed using noiseless theoretical waveforms of BNSs and BBHs to answer the question: 'can a ML algorithm correctly distinguish between BNS and BBH waveforms?' Third, classifier predictions are visualized at the level of individual time-series signals. Finally, the limits of machine-learning generalization are explored using cross-source tests in mass space.

Cell 1: Classical Matched Filtering with Real LIGO Data

The first cell reproduces the matched-filter analysis used in the GWOSC LOSC tutorial for the binary black hole event GW150914. The calibrated strain data from the detector are modeled as

$$s(t) = h(t;\theta) + n(t), \tag{21}$$

where $h(t;\theta)$ is the GW signal predicted by general relativity for source parameters θ, and $n(t)$ is detector noise that is approximately stationary and Gaussian.

The strain data are conditioned using standard LIGO procedures: removal of edge artifacts, high-pass filtering above 30 Hz, and resampling to a uniform cadence. The noise power spectral density (PSD) is estimated directly from the data and defines the frequency-dependent noise variance.

Matched filtering is a statistical technique to find the best matching theoretical waveform, i.e., a template, for the signal in your data. While full matched filtering calculations are computationally expensive due to needing to search through large banks of templates, for demonstration here we use a single GW template with the parameters of the oberved source GW150914. Given a strain time series $s(t)$ and a template waveform $h(t)$, the matched-filter signal-to-noise ratio (SNR) as a function of time is defined by

$$\rho(t) = \frac{(s|h_t)}{\sqrt{(h|h)}}, \tag{22}$$

where $h_t(t') \equiv h(t'-t)$ denotes the template shifted by time t, and

$$(h|h) = 4\int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\,df \tag{23}$$

is the optimal SNR of the source. $S_n(f)$ is the PSD of the noise. The resulting SNR time series peaks when the template best matches the data, providing an estimate of the merger time and the detection significance.

This section establishes the role of accurate physical modeling, explicit noise assumptions, and optimal detection statistics in GW astronomy.

Cell 2: Machine Learning Classification of BBH versus BNS Waveforms

The second cell addresses a problem of classification. Rather than asking whether a signal is present, the task is to determine whether a given GW signal originates from a BBH or a BNS.

The dataset consists of noiseless theoretical waveforms generated with PyCBC. In the first demo, the GWs from BNSs and BBHs are modeled with post-Newtonian inspiral waveforms at a point in the binary evolution well before the merger. We have a version that also uses full inspiral-merger-ringdown waveform models, in which case the final one second of the waveform is extracted, ensuring that the late inspiral and merger—where ground-based detectors are most sensitive—are included.

All waveforms are time-aligned and represented as fixed-length vectors. This deliberately removes extrinsic complications and isolates intrinsic waveform morphology. The strain amplitudes are log-scaled to stabilize the dynamic range, and the data are standardized before dimensionality reduction.

Principal component analysis (PCA) is applied to identify the dominant modes of waveform variation. Physically, the leading components encode mass-dependent features such as chirp rate, merger frequency, and ringdown decay time. Only a small number of components capture most of the variance, indicating that the relevant physics is low dimensional.

A logistic regression classifier trained on these features achieves near-perfect precision and recall. This performance reflects the strong physical separability of BBH and BNS waveforms in the late inspiral and merger regime, rather than model complexity.

Cell 3: Visualization of True and Predicted Labels

The third cell provides a qualitative check on classifier behavior. A subset of test waveforms is plotted in the time domain, with annotations indicating both the true physical label and the predicted label.

Each panel overlays identical strain data with different line styles to represent the true and predicted classifications. This visualization emphasizes that the classifier assigns interpretations based on learned morphology rather than modifying the waveform itself. The agreement between true and predicted labels reinforces that the classification is driven by physically meaningful differences in signal structure.

Cell 4: Cross-source Generalization

The final cell examines the limits of machine learning by testing generalization across disjoint mass ranges. The classifier is trained on one class of sources, such as low-mass BBHs and BNSs, and evaluated on a separate class with non-overlapping, higher masses, or vice versa.

All preprocessing steps are fitted exclusively on the training data to prevent information leakage. Despite identical waveform generation and analysis procedures, classification performance degrades sharply. In extreme cases, the classifier labels all test signals as one class, yielding an accuracy no better than random guessing.

This result illustrates a fundamental limitation of standard machine-learning models: they interpolate well within the training distribution but extrapolate poorly beyond it. Physically, the classifier learns correlations tied to specific mass scales rather than universal features of compact binary coalescence.

Homework Exercise # 5: redo this computational demo, but replace the inspiral-only waveform with the full (inspiral+merger+ringdown) waveform.

Conclusions

This notebook demonstrates that matched filtering and machine learning address fundamentally different questions. Matched filtering is a detection strategy rooted in explicit physical models and noise statistics, while machine learning exploits waveform morphology within the domain represented in the training data.

Used together, these approaches clarify both the physical content of GW signals and the limitations of data-driven methods. In particular, the cross-source tests highlight the necessity of physically motivated modeling and representative datasets when applying machine learning in GW astronomy.